

Date:

3.

(a) Given,

$$\text{Cache} = 2 \text{KB} = 2^1 \times 2^{10} \text{ Bytes} = 2^{11} \text{ Bytes}$$

$$\begin{aligned} \text{Data per block} &= 8 \text{ words} = 2^3 \text{ words} \rightarrow 2^m \\ &= 8 \times 2 \text{ bytes} = 16 \text{ bytes} = 2^4 \text{ Bytes} \end{aligned}$$

$$\text{No of block in cache} = \frac{\text{Cache size}}{\text{Data per block}} \quad \left[\begin{array}{l} 1 \text{ word} = 2 \text{ byte} \\ = 16 \text{ bit} \end{array} \right]$$

$$= \frac{2^{11}}{2^4}$$

$$= 2^7$$

$$\therefore n = 7$$

$$m = 3$$

$$\text{Actual size of cache} = 2^n \times (2^m \times 16 + 64 - (n + m + 1) + 1)$$

$$= 2^7 \times (2^3 \times 16 + 64 - (7 + 3 + 1) + 1)$$

$$= 23296 \text{ bits} //$$

b.

80	Miss
96	Miss
120	Miss
88	Hit
16	Miss
96	Hit

0	
1	[16-31]
5	[80-95]
6	[96-111]
7	[112-127]

0-15
16-31
32-47
48-63
64-79
80-95
96-111
112-127

80:

$$\text{Block address} = \left\lfloor \frac{80}{16} \right\rfloor = 5$$

$$\text{Block number} = 5 \times 2^7 = 5$$

Miss.

[80-95] copied

96:

$$\text{Block address} = \left\lfloor \frac{96}{16} \right\rfloor = 6$$

$$\text{Block number} = 6 \times 2^7 = 6$$

Miss.

[96-111] copied

120:

$$\text{Block address} = \left\lfloor \frac{120}{16} \right\rfloor = 7$$

$$\text{Block number} = 7 \times 2^7 = 7$$

Miss.

[112-127] copied

88:

$$\text{Block address} = \left\lfloor \frac{88}{16} \right\rfloor = 5$$

$$\text{Block number} = 5 \times 2^7 = 5$$

Hit

16:

$$\text{Block address} = \left\lfloor \frac{16}{16} \right\rfloor = 1$$

$$\text{Block number} = 1 \times 2^7 = 1$$

Miss

[16-31] copied

96:

$$\text{Block address} = \left\lfloor \frac{96}{16} \right\rfloor = 6$$

$$\text{Block number} = 6 \times 2^7 = 6$$

Hit.

$$\text{Miss ratio} = \frac{\text{Miss}}{\text{Total Access}}$$

$$= \frac{4}{6} = 0.67$$

$$= 66.67\%$$

ii) using forwarding: 12 cycle $\times$ 130 ns

iii) A.

Before loop: 3 ins

Loop iteration: 6 times.

Inside loop: total 9 instruction.

but 4 times: 9 instruction

2 times: 8 instruction

After loop: 1 ins.

$$\begin{aligned}\text{Total} &= 3 + 4 \times 9 + 2 \times 8 + 1 \\ &= 3 + 36 + 16 + 1 \\ &= 56 \text{ instruction.}\end{aligned}$$

in 7th iteration: condⁿ check. so 1 ins.

$$\therefore \text{Total} = 57 \text{ instr}$$

$$\text{time} = 57 \times T_c$$

B. (lw-slt) $\rightarrow$ 1 stall [loop runs for 6 times]

$$\therefore 6 \times 1 \text{ (load-use stall)}$$

j taken $\rightarrow$ 6 times $\therefore$ 6 stall for jump

b.eq taken 1 time (at 7th iter), bne taken 2 times:

$$\text{total} = 6 + 6 + 2 + 1 = 15 \text{ stalls.}$$